25ECPCC302IAE-I

Seat No. EM25705S

15 AUG 2026

Student Name : Harsh

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, August 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC302 - SDC



Time : 1 Hour

Maximum Marks : 20

Instructions

- All Questions are compulsory and carry equal weightage of 1 mark
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	BL
	a)Illustrate the difference between CE, CB,CC configuration	1	CO1	L1
	b)Discuss the role of emitter	1	CO1	L3
	c)Why is a MOSFET considered a voltage-controlled device	1	CO2	L2
	d)State the difference between BJT and FET	1	CO2	L2
	e)What does LED stand for?	1	CO3	L2
	f)Define a photodiode	1	CO3	L2
Q2	a)Explain the creation of depletion region and effect of forward biasing	5	CO1	L2
	OR			
	b)Derive the expression for maximum values of operating point Ic and Vce	5	CO1	L2
Q3	a)List down the Applications of MOSFET in detail	5	CO2	L2
	OR			
	b)Explain the working of N-channel JFET with proper diagram	5	CO2	L3
Q4	a)Explain the principle of emerging semiconductor device SET	5	CO3	L2
	OR			
	b)Discuss the construction of Photovoltic cell in detail	5	CO3	L3